

Trace LiBF₄ Enabling Robust LiF-Rich Interphase for Durable Low-Temperature Lithium-Ion Pouch Cells

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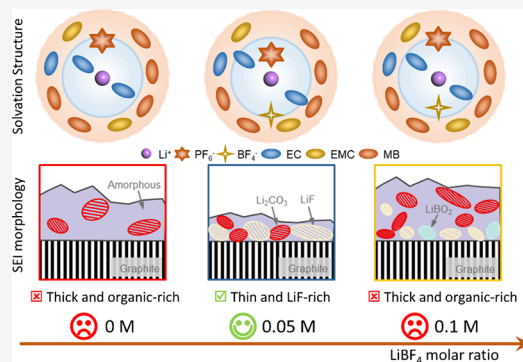


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Supporting Information

ABSTRACT: Lithium-ion batteries (LIBs) with electrolytes containing lithium tetrafluoroborate (LiBF₄) can achieve large capacity retention under low temperature, but the accompanying rapid capacity decay inhibits commercialization. Here, the impact of LiBF₄ as a supplemental salt to LiPF₆ is systematically investigated using low ethylene carbonate (EC)-content electrolytes, along with a low-melting-point cosolvent. It is found that rational adjustment of the amount of LiBF₄ could not only regulate the interactions of anions and solvents in Li⁺ solvation sheaths but also tune the composition and morphology of solid electrolyte interphase (SEI). It is worth noting that electrolytes with trace amount of LiBF₄ (0.05 M) show synergistic interaction between PF₆[−] and Li⁺ and decreased interaction between EC and Li⁺, achieving a dense and LiF-rich SEI, which enables a 200 mAh pouch cell with less gas generation, long-lived cycling, and higher low-temperature capacity, simultaneously. This work provides new insight into utilizing trace LiBF₄ for stable interface construction of durable low-temperature LIBs.



Lithium-ion batteries (LIB) with layered oxide cathodes, graphite anode, and carbonate electrolyte have been successfully commercialized and utilized as an important electrochemical energy storage device in a wide spectrum of application scenarios, including portable electronics, electric vehicles, and grid-scale energy storage.^{1,2} The main solvent, ethylene carbonate (EC), shows indispensable function for constructing stable solid electrolyte interphase (SEI) on graphite to ensure long-lived cycling.³ However, EC greatly restricts the low-temperature application of LIBs due to its high melting point (36.4 °C), strong solvation with Li ions, and formation of highly resistive SEI.⁴ Sluggish interfacial kinetics are exhibited under low temperatures due to electrolyte freezing, high impedance for desolvation, and Li-ion transport through SEI.⁵ The deteriorated interface causes a large voltage polarization, resulting in low capacity and potentially Li plating on graphite, endangering the stability and safety of LIBs at low temperatures.^{6,7} To regulate the interfacial composition and kinetics, formulation of new electrolytes to improve the lithium ion transport properties at low temperatures is the most common and crucial approach.⁸

Since the deteriorated lithium-ion transport at low temperatures originates from the strong interaction between EC and

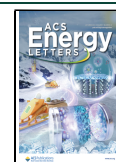
Li⁺, one ultimate solution is to develop a new electrolyte system without EC. To maintain effective SEI formation on graphite, various EC-free electrolytes have been studied using alternative lithium salt, such as Li bis(fluorosulfonyl)imide (LiFSI), or film formation additives such as fluoroethylene carbonate (FEC) or vinylene carbonate (VC).^{9–12} With the help of low melting solvents and/or diluents, these EC-free electrolytes form a LiF-rich SEI contributing to low interfacial resistance at low temperatures, which enables high-performance low-temperature LIBs with superior capacity and cycling stability.^{13–17} However, the low purity and continuous corrosion of the Al current collector due to the insufficient passivation capability of new salts as well as the high cost of fluorinated solvents still hinder the commercialization of these electrolytes.¹⁸ The other approach is to reduce the usage of EC by adding a low-melting-point cosolvent including linear esters

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like methyl acetate (MA), methyl propionate (MP), and methyl butyrate (MB) or linear carbonates like ethyl methyl carbonate (EMC).^{19–22} Since 2000, various works have been reported to optimize low-EC-content electrolytes by adding inorganic or organic additives to improve the performance of LIBs at low temperatures.^{23–26} Among the latter solutions, the use of lithium tetrafluoroborate (LiBF_4) in electrolytes can effectively achieve a lower charge transfer resistance than single-salt electrolytes using LiPF_6 , realizing high capacity retention at subfreezing temperatures. Therefore, LiBF_4 can be a promising salt additive for low-temperature LIBs.²⁷ On the other hand, LIBs using electrolytes containing a high level of LiBF_4 , higher than 0.1 M, generally suffer from an accelerated capacity decay during normal operations. The accelerated performance decay is generally attributed to a poor interfacial passivation on graphite.^{3,8,28} Why LiBF_4 shows poor a passivation property and how the synergy effects of LiBF_4 with other anions on solvation structure, interfacial composition, and kinetics to achieve high stability and low-temperature capacity have yet to be clarified.³

In this work, a low-EC-content electrolyte with various LiBF_4 amounts is systematically investigated to understand the effect of LiBF_4 on the solvation structure, interfacial composition, and resistances as well as pouch cell performance. The strong interaction of BF_4^- with Li^+ is demonstrated to effectively regulate the solvation structure by influencing the interaction between PF_6^- and Li^+ as well as EC with Li^+ . Based on combined theoretical calculation and characterization, electrolytes with more than 0.1 M LiBF_4 exhibit strong affinity between BF_4^- with Li^+ , forming a poor protective interface induced by more BF_4^- decomposition, which leads to continual solvent decomposition. For comparison, electrolytes with trace LiBF_4 (0.05 M) enable increased coordination between PF_6^- and Li^+ and also decreased interaction between EC and Li^+ , forming an effective passivated and LiF-rich interface by more PF_6^- decomposition. By employing a 200 mAh $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$ ||graphite (NMC532||Gr) pouch cell, this interface formed in electrolytes with trace LiBF_4 realizes stable interfacial stability and low interfacial resistance, simultaneously showing less gas during formation, low self-discharge, long-lived cycling, and large low-temperature capacity. This work offers insight into the intertwined effects of BF_4^- on solvation structure and SEI structure and also a guideline for designing electrolytes of low-temperature lithium-ion batteries with high stability.

Properties of Electrolytes with Various LiBF_4 Ratios.

The low-EC-content ester electrolyte is used as the baseline electrolyte to study the effect of LiBF_4 on interfacial properties. MB with low melting point of -85.8°C is used as cosolvent, and the formulation of base low-EC-content electrolyte is 1 M LiPF_6 in EC-EMC-MB (2:2:6 by volume), denoted as E_0 . Further, 0.05, 0.1, and 0.2 M LiBF_4 were added in E_0 to generate three electrolytes for further study, denoted as E_1 , E_2 , and E_3 , respectively.

Redox activities of electrolytes were studied using density functional theory (DFT) calculations, which are vital factors for different interfacial compositions and kinetics. Lowest unoccupied molecular orbital (LUMO) energy levels of solvents and salts were calculated by DFT and compared in Figure 1a and Table S1. The salt molecules exhibit a lower LUMO energy level than the solvent molecules, indicating a greater accessibility to decomposition on the anode surface. The anions are typically reduced at higher reduction potential

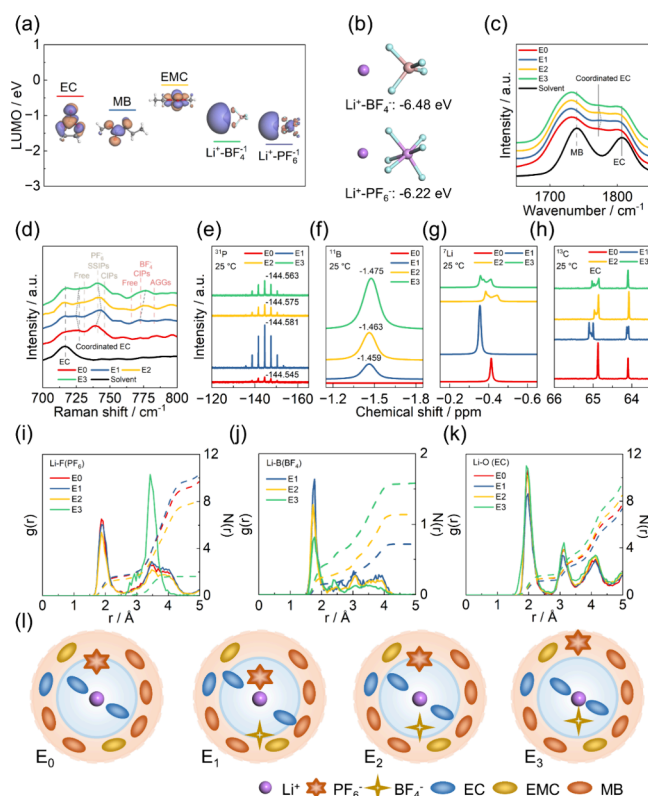


Figure 1. (a) Lowest unoccupied molecular orbital (LUMO) energy levels of EC, MB, EMC, $\text{Li}^+\text{-BF}_4^-$, and $\text{Li}^+\text{-PF}_6^-$ obtained by DFT calculations. (b) Binding energies of Li^+ -anion complexes. (c) Fourier transform infrared spectroscopy (FTIR) spectra of the C=O bond in electrolytes and base solvent. (d) Raman spectra of electrolytes and base solvent. (e–h) ^{31}P , ^{11}B , ^7Li , and ^{13}C nuclear magnetic resonance (NMR) spectra of electrolytes under 25°C . (i–k) Radial distribution functions and coordination number of electrolytes from MD simulation. (l) Schematics illustrating solvation structure.

above 1 V vs Li/Li^+ , producing a LiF-rich solid electrolyte interphase (SEI) with stable chemical and mechanical stability. However, for routine ester electrolyte, the solvents in the inner solvation sheath would be preferentially reduced below 0.8 V vs Li/Li^+ , forming a highly resistive SEI and resulting in sluggish kinetics.^{29,30} The binding energies of Li^+ -anion complexes were further calculated by DFT. As revealed in Figure 1b, the binding energy of BF_4^- with Li^+ is larger than that of PF_6^- with Li^+ , indicating a stronger force in BF_4^- - Li^+ compared to PF_6^- - Li^+ . The enhanced interaction between Li^+ and anions could enable Li^+ to more easily attract electrons from anions, leading to the formation of more anion-derived SEI for fast kinetics and long-lived cycling.^{31,32}

To probe the solvation structure of electrolytes, Fourier transform infrared spectroscopy (FTIR) and Raman analysis were conducted. As shown in Figures 1c and S1, typical stretching vibration peaks of C=O located at 1805 and 1739 cm^{-1} are observed, which are ascribed to the C=O in EC and MB.^{33,34} New peaks emerged at $\sim 1773\text{ cm}^{-1}$ compared to pure EC, corresponding to coordinated EC with Li^+ . The C=O peaks of coordinated EC shift from $\sim 1773\text{ cm}^{-1}$ in E_0 to higher wavelength of $\sim 1776\text{ cm}^{-1}$ in E_1 , which implied that weaker interaction between Li^+ and EC in E_1 . The C=O peaks of coordinated EC shift to lower wavelength of ~ 1774 and $\sim 1771\text{ cm}^{-1}$ in E_2 and E_3 , implying interaction between Li^+

and EC become stronger upon adding LiBF₄ amount more than 0.1 M. For comparison, the C=O in MB in different electrolytes shows similar wavelength. The solvation structure was further studied by Raman analysis in Figure 1d. In E₀, EC occupies the inner solvation sheath due to the high coordination capability, while PF₆[−] shows weak interaction to Li⁺ with a peak at 739 cm^{−1}, corresponding to mainly a solvent-shared ion pair (SSIP). The Raman shifts of coordinated EC in different electrolytes show good consistency with the peak wavelength in FTIR curves, where E₁ exhibits the lowest Raman shifts of 725 cm^{−1}, implying the weakest interaction between Li⁺ and EC in E₁. Moreover, PF₆[−] exhibits simultaneously the highest Raman shifts of 744 cm^{−1} in E₁ compared to 742 and 741 cm^{−1} in E₂ and E₃, respectively, indicating a greater amount of PF₆[−] was coordinated in E₁.^{35–37} The Raman shift of BF₄[−] moves to a higher value with increasing LiBF₄ ratio, indicating the BF₄[−] molecules take part in the solvation shell and form more contact ion pairs (CIPs) in E₂ and E₃.^{38,39} Hence, it is confirmed that the introduction of LiBF₄ could regulate the solvation structure.

Nuclear magnetic resonance (NMR) measurements were conducted to further investigate the solvation structure in Figure 1e–h. The ³¹P NMR peak of E₁ shows the lowest value compared to other electrolytes, while a trend of downfield (positive) shift occurs upon adding more LiBF₄. The more negative ³¹P indicates a decreased electron density in PF₆[−], verifying an enhanced PF₆[−]–Li⁺ ion pairing in E₁. Also, upfield chemical shifts of ¹¹B-NMR peaks indicate decreased electron density in BF₄[−], verifying stronger interaction between BF₄[−] and Li ion in electrolytes with a greater amount of LiBF₄.⁴⁰ Moreover, contrary to the sharp and single resonance of the ⁷Li NMR peak in E₀ and E₁, asymmetric shapes are observed in the ⁷Li NMR peak in E₂ and E₃, suggesting the presence of a second lithium species in electrolytes with a greater amount of LiBF₄.³⁵ Importantly, E₁ exhibits a more positive value with an increased electron density around Li⁺, whereas there are upfield shifts of ⁷Li NMR peak in E₂ and E₃, which are due to either an increased ion pairing of BF₄[−] or a stronger ion solvation. The upfield shifts of ¹³C NMR of E₂ and E₃ support the stronger ion solvation as adding more LiBF₄, which might be due to the increased Li-ion mole ratio. Similar NMR results are obtained upon testing under −20 °C (as shown in Figure S2), indicating that different solvation structures are retained under low temperature. Overall, a trace amount of LiBF₄ in E₁ enables stronger interaction of PF₆[−] with Li ions yet a lower ion solvation ability and similar solvation structure with E₀, which could not be realized in electrolytes with LiBF₄ amount greater than 0.1 M.

The solvation structure is further quantified by molecular dynamics (MD) simulations. Radial distribution functions (RDFs) show a sharp peak of Li–F_{PF₆}, Li–B_{BF₄}, and Li–O_{EC} within 3 Å, confirming that PF₆[−], BF₄[−], and EC enter the inner solvation shell in E₀–E₂. The RDF values for Li–F_{PF₆}, Li–B_{BF₄}, and Li–O_{EC} are ~1.92, 1.77, and 1.98 Å, respectively, which agree well with preferential reduction of anions during lithiation. Importantly, E₁ shows a highest time-averaged coordination number (CN) of Li–F_{PF₆} and a lowest CN of Li–B_{BF₄} and Li–O_{EC} among all electrolytes, which are 1.7 F_{PF₆}, 0.18 B_{BF₄}, and 1.21 O_{EC} per Li⁺, confirming the solvation structure analyzed by characterizations. In contrast, along with enhanced CN of Li–B_{BF₄} in E₂ and E₃ (0.31 and 0.6), CN of

Li–F_{PF₆} decreases and CN of Li–O_{EC} increases, where the RDF of Li–F_{PF₆} in E₃ even increased to 3.425 Å, verifying inversely increased coordinated solvents for electrolytes adding more LiBF₄ than E₁, which could be due to increased amounts of Li⁺ in the same volume of solvents. Therefore, the solvation structure is confirmed to be well-regulated in the electrolyte with adding LiBF₄ (illustrated in Figure 11), which might be attributed to stronger Li⁺–BF₄[−] interaction weakening the spatial shielding effect of solvents, contributing to more PF₆[−] entering the inner solvation sheath.⁴¹ The electrolyte E₁ with 0.05 M LiBF₄ exhibits increased coordination of PF₆[−] as well as the weakest solvation of EC and coordination of BF₄[−], which potentially constructs a stable inorganic-rich SEI by more PF₆[−] reduction and exhibits low desolvation impedance.

Electrochemical Properties in Initial Formation Cycles of Graphite Anodes. To further testify to the effect of solvation affinity in Li⁺–solvents and Li⁺–anions on interfacial formation, initial formation cycles of graphite electrodes were investigated where both anion and solvent reduction occur during the initial lithiation. Here, single-channel three-electrode cells were used to accurately monitor and control the graphite electrode potential.⁴² As shown in Figures 2a and

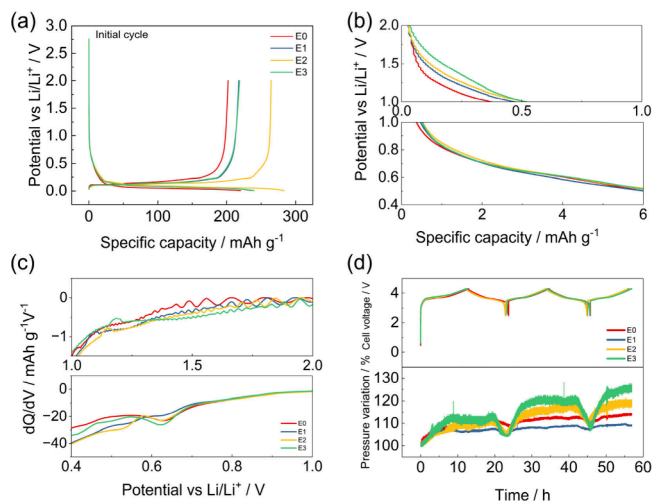


Figure 2. (a) Potential curves during the initial cycle. (b) Enlarged initial-cycle potential curves correspond to anion and solvent decomposition. (c) dQ/dV curves of electrolyte decomposition in the initial cycle. (d) In situ cell pressure changes during the initial three charge–discharge cycles.

S4, different initial discharge capacities are observed, where E₂ shows initial larger capacity than electrolytes with a smaller amount of LiBF₄ (E₀ and E₁). The anion and solvent decompositions were studied in detail in enlarged initial potential curves and differential capacity curves (dQ/dV) in Figure 2b,c, which reveal that SEI formation is contributed to anion reduction at ~1.5 V and solvent reduction at ~0.65 V vs Li/Li⁺. Larger capacities are consumed and larger plateaus are observed above 1 V in E₁–E₃ compared to E₀, which verifies more anions were reduced, corresponding well to the enhanced coordination of anions by adding LiBF₄ (as illustrated in Figure 1). Moreover, E₁ exhibits the lowest dQ/dV value of plateau at ~0.65 V, which shows the weakest solvent reduction compared to others, implying a more stable SEI was formed by anions to avoid severe solvent decomposition. For comparison, E₂ and E₃ show stronger

solvent decomposition. E_3 exhibits an even larger dQ/dV value than E_0 . These results imply the reduction products of LiBF_4 hinder the formation of a dense protective film, leading to continual decomposition of the solvent, which agrees with the reported poor anode passivation property of LiBF_4 .⁴³ The results indicate stable anion-derived SEIs were formed in E_1 , which establish stable interfacial protection to avoid severe solvent reduction, benefiting from more PF_6^- being coordinated that would be preferentially reduced.

Stimulated by a coordinated anion-rich solvation structure and exceptional electrochemical interfacial passivation in E_1 , pressure variations of pouch cells were measured to further validate the reduced solvent decomposition of E_1 . The 200 mAh $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2\|\text{graphite}$ (NMC532||Gr) 402035 pouch cells with the same cell volume were cycled coupled with in situ stacking pressure monitoring using film pressure sensors. As shown in Figure 2d, during charging and discharging under 0.1 C, the detected cell pressures all increase upon cycling, attributed to cell volume expansion by the generated gas during solvent decomposition.¹⁰ As expected, after the second charging process, the pouch cell with E_1 shows a lower pressure variation compared to the pouch cell with E_0 , of which the pressure increase 7.49% and 11.62% for E_1 and E_0 , respectively. In sharp contrast, pouch cells with E_2 and E_3 exhibits severe pressure increase of 15.87% and 20.53%, respectively, proving the poor SEI passivation of electrolytes with large amount of LiBF_4 . Meanwhile, similar pressure variations between second and third charging processes are observed for pouch cells with E_0 – E_2 , while pouch cell with E_3 still displays a large pressure increase of 25.05% after third charging. Moreover, the cell volume changes were also evaluated by storing the cycled pouch cells at ambient temperature for 1 month, as shown in Figure S5. The pouch cell with E_3 shows substantial swelling, which fits well with the greatly increased cell stack pressure. As a comparison, gas production was negligible in the pouch cells with E_0 – E_2 . In short, the efficient protective SEI formed in E_1 shows outstanding merits in interfacial stability.

Composition and Structure of Anion-Derived SEIs.

The morphologies of anion-derived SEIs were evaluated using transmission electron microscopy (TEM) to determine the effect of solvation structure on anion reduction, further verifying the observed reduced solvent decomposition in E_1 . The graphite electrodes were lithiated in E_0 – E_2 electrolytes from open circuit potential to 1 V vs Li/Li^+ to avoid solvent decomposition. As shown in Figure 3a, the potential curves of graphite electrodes in E_1 and E_2 show obvious peaks around 1.5 V vs Li/Li^+ , corresponding to the reduction of LiPF_6 and LiBF_4 . Similar to Figure 2a–c, E_1 and E_2 exhibit larger capacity of 8.33 and 8.57 mAh g^{-1} in comparison to that of 7.35 mAh g^{-1} in E_0 , implying more anions were reduced and a large amount of SEI was accumulated. The graphite electrode harvested from E_0 shows mainly a clean graphite layer in Figure 3b, where some amorphous regions dotted with few crystalline particles are grown on the edge of the graphite. As enlarged in Figure 3c–e, the dark particles are small LiF crystals with a size of ~ 2 nm, proving the limited decomposition of PF_6^- in E_0 . For comparison, a greater amount of decomposition products is observed on graphite in E_1 and E_2 , which exhibits larger contrast in TEM images in Figure 3g,m. On graphite in E_1 (Figure 3h), some continuous films are coated, in which various inorganic nanoparticles with a size of ~ 4 nm could be observed. As enlarged in Figure 3i–k,

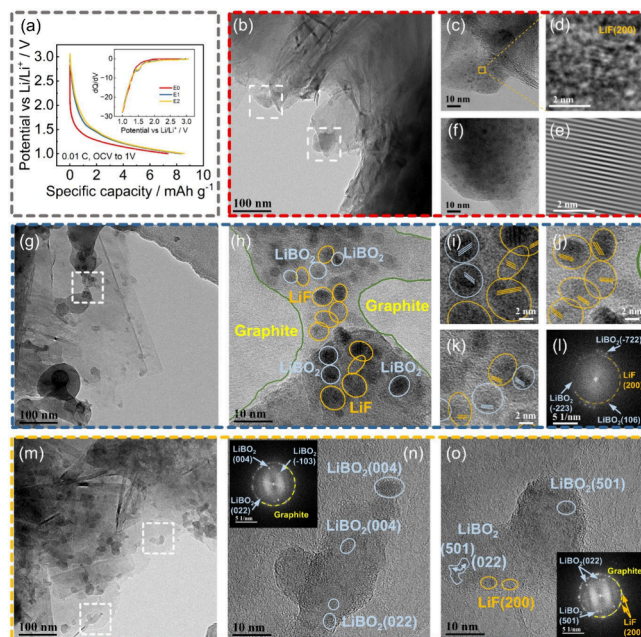


Figure 3. Anion-derived SEIs on graphite in different electrolytes. (a) Potential curves during lithiation to 1 V vs Li/Li^+ and corresponding dQ/dV curves. Low- and high-magnification TEM images of graphite electrodes after lithiation to 1 V vs Li/Li^+ in E_0 (b–e), E_1 (g–l), and E_2 (m–o).

the inorganic species are LiF and LiBO_2 , as supported by lattice spacing and the corresponding fast Fourier transform (FFT) image (Figure 3l), verifying the enhanced decomposition of anions in E_1 . In E_2 , the inorganic species are mainly LiBO_2 with larger size and low amount (Figures 3m–o and S7), which could not form a continuous protection film to prevent solvent reduction, illustrating the poor passivating property upon adding a greater amount of LiBF_4 . These findings coincide well with the solvation structure revealed in Figure 1, where electrolytes with trace LiBF_4 (0.05 M) show enhanced coordination of PF_6^- yet decreased interactions between EC and Li^+ . Therefore, a continuous LiF -rich SEI by enhancing the reduction of PF_6^- could be obtained through solvation structure regulation in E_1 .

Morphologies and composition of SEIs on graphite anodes after formation cycles. Morphologies and components of SEIs after cycle formation were further evaluated using TEM. The SEI shows a thinner thickness of 3.4–11 nm in E_1 compared to SEIs in E_0 and E_2 , as shown in Figure 4a–c and Figures S8–S10. The thinner SEI in E_1 possesses a dense inorganic-layer including various LiF , Li_2CO_3 , and LiBO_2 nanoparticles, which favors Li -ion diffusion due to short immigration length and aggressive synergetic effects of abundant inorganic contact interfaces. Also, this LiF -rich SEI has the potential to enable low graphite interfacial resistance at low temperature.¹³ For comparison, mainly Li_2CO_3 islands are randomly dispersed in the amorphous SEI in E_2 , which originate from solvent decomposition (more positions in Figure S10). The discrepancy in the thickness and composition further validates suppressed solvent decomposition in E_1 , which is strongly related to initial anion decomposition products as revealed in Figure 3.

Surface chemistry was further characterized by using X-ray photoelectron spectroscopy (XPS) measurements. Basically, the atomic concentrations are compared in Figure 4d, which

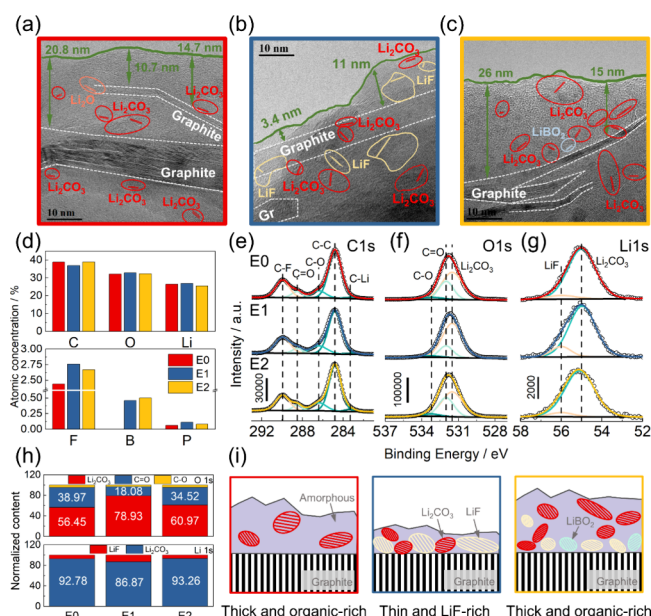


Figure 4. Composition and structure of SEI on graphite electrodes after three cycles. (a–c) TEM images of SEI on graphite in electrolyte E₀–E₂. (d) Atomic concentration of elements in the SEI. (e–g) C 1s, F 1s, and Li 1s spectra of graphite electrodes after 3 cycles in different electrolyte. (h) Normalized contents of Li₂CO₃, C=O and C–O in O 1s spectra as well as LiF and Li₂CO₃ from Li 1s spectra. (i) Schematic illustration of SEIs formed on graphite in different electrolyte.

show much difference among the three electrolytes. Compared to E₀ and E₂, the SEI formed in E₁ shows the lowest C atomic ratio but highest O, Li, and F atomic ratios, indicating a lower proportion of organic species but a high fraction of inorganic species for the SEI formed in the E₁ electrolyte. In addition, the E₁ electrolyte generates a high ratio of P but lesser ratio of B than E₂, which originates from PF₆[−] and BF₄[−] decomposition, confirming more PF₆[−] participates in the inner

solvation sheath and decomposed. XPS spectra were deconvoluted to reveal the detailed components, as shown in Figure 4e–g and Figures S11 and S12. As shown in the C 1s spectra, E₁ shows lower C=O and C–O peaks than E₀ and E₂, ascribed to the lowest organic component. A similar low proportion of C=O and C–O species is also confirmed in O 1s spectra. Meanwhile, the normalized content calculated based on the peak area in O 1s and Li 1s are summarized in Figure 4h. More Li₂CO₃ compared to organic species and more LiF than Li₂CO₃ is detected in the SEI formed on graphite cycled in E₁ electrolyte. This is consistent with the morphologies and components observed in the TEM images. It is noteworthy that compared to SEI formed in E₂, E₁ exhibits more Li_xPO_yF_z but less B–O in P 2p and B 1s spectra, which are products of LiPF₆ and LiBF₄ decomposition, respectively. This is consistent with the formation of solid electrolyte interphase (SEI) influenced by solvation structure, where more LiPF₆ is reduced in E₁, while more LiBF₄ is reduced in E₂. Comprehensively, schematic illustration of SEIs formed in different electrolytes is shown in Figure 4i. E₁ electrolyte facilitates the SEI with thin thickness, abundant phase boundaries, and dense LiF-rich layer. For comparison, E₂ electrolyte with higher amount of BF₄[−] forms a final thick SEI from severe solvent decomposition, attributed to more reduction products (LiBO₂) from BF₄[−] hindering the continuity of the protective LiF-rich layer. Therefore, such a highly protective and LiF-rich layer obtained by E₁ would further contribute to high kinetics and interfacial stability.

Interfacial Resistances and Kinetics. The interfacial resistances of graphite electrodes in pouch cells were decoupled via a three-electrode testing procedure to evaluate the influence of solvation structure and SEI on the interfacial electrochemical processes.⁴² Here, the multitemperature electrochemical impedance spectroscopy (EIS) tests were conducted in a pouch cell to study the actual impedance in a full cell and also avoid the interference of large contact resistance in the coin cell (Figure S13). Distribution of

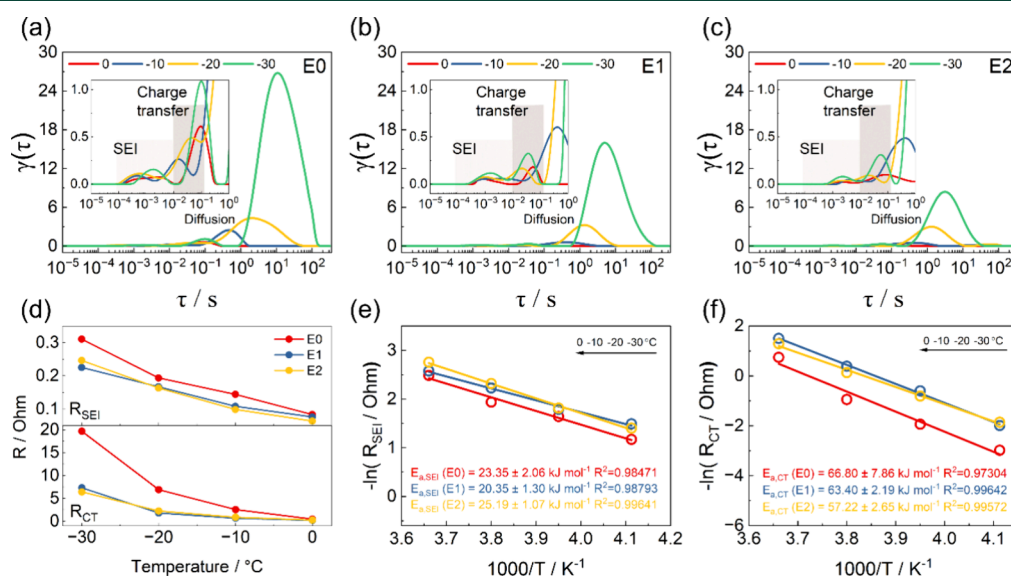


Figure 5. Interfacial kinetics of Li⁺ intercalation into graphite electrode under low temperature. (a–c) DRT transformation of multitemperature EIS results of graphite electrodes in pouch cells. (d) Composition of interfacial resistance of graphite in different electrolytes. (e and f) Activation energy barriers of Li-ion transport process through SEI and charge transfer process by fitting EIS resistance according to Arrhenius law.

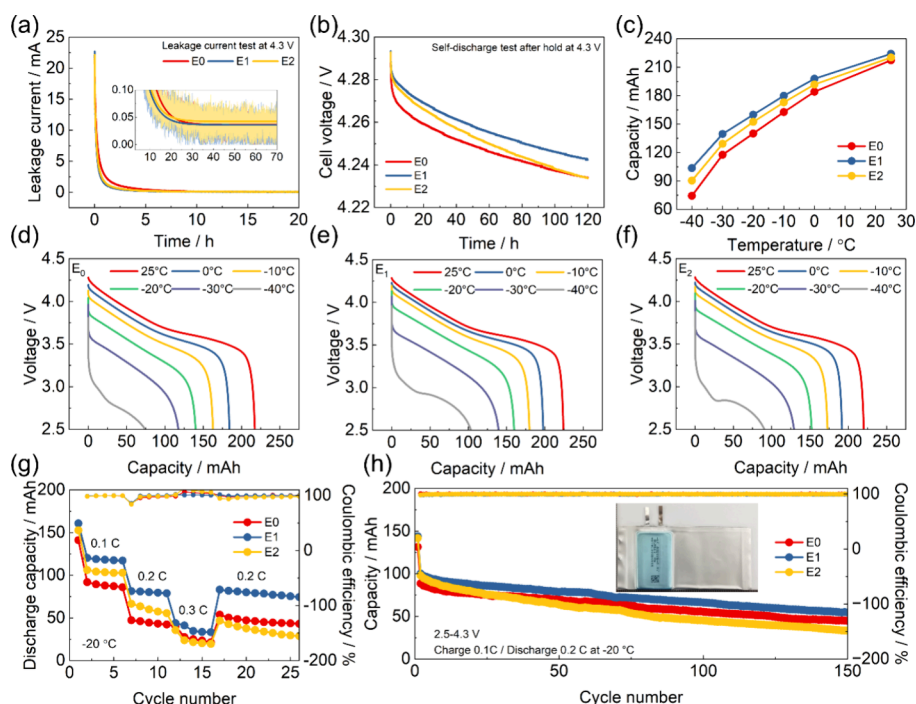


Figure 6. Interfacial stability and low-temperature performance of a 200 mAh NCM532||Gr pouch cell. (a) Leakage current curves tested by a potentiostatic hold at 4.3 V for 20 h. (b) Self-discharge voltage results during 120 h rest after CCCV was charged to 4.3 V. (c) Low-temperature discharge capacity after room-temperature charge. (d–f) Discharge potential curves at different temperature. (g) Rate performance under -20°C . (h) Cycle performance under -20°C at 0.1 C charge and 0.2 C discharge.

relaxation time (DRT) was used to distinguish resistance evolution of specific electrochemical processes. The DRT results are compared in Figure 5a–c with three apparent peaks in time scale, by which overlapped semicircles due to closed characteristic frequency were decoupled.^{44,45} Generally, these three peaks located at 10^{-4} to 10^{-2} , 10^{-2} to 10^{-1} , and 10^{-1} to 10 s are ascribed to Li^+ transport through SEI (R_{SEI}), charge transfer (R_{CT}), and diffusion. E_1 and E_2 show a lower R_{SEI} value of 0.07 and 0.04 $\Omega\text{m}_{(-20^{\circ}\text{C})}$ than 0.11 $\Omega\text{m}_{(-20^{\circ}\text{C})}$ of E_0 , indicating a faster Li^+ migration in the inorganic-rich SEI in E_1 and E_2 . Meanwhile, the R_{CT} value of 0.16 and 0.09 $\Omega\text{m}_{(-20^{\circ}\text{C})}$ in E_1 and E_2 are largely decreased compare to 0.49 $\Omega\text{m}_{(-20^{\circ}\text{C})}$ in E_0 , indicating a fast desolvation process, which is consistent to the weak interaction between EC and Li^+ in E_1 and E_2 . Similar results were also obtained by comparing the resistance of interfacial processes by fitting the EIS curves based on the frequency region identified by DRT, as shown in Figure 5d and Tables S2 and S3. Activation energies of Li^+ transport in SEI ($E_{\text{a,SEI}}$) and charge transfer ($E_{\text{a,CT}}$) were calculated by fitting resistance results using the Arrhenius equation. As shown in Figure 5e, the $E_{\text{a,SEI}}$ values of E_0 – E_2 are 23.35, 20.35, and 25.19 kJ mol^{-1} , respectively, implying rapid Li^+ diffusion in the SEI formed in E_1 , resulting from the thin thickness of SEI and aggressive synergetic effects of abundant inorganic contact boundaries. It is also noticed that $E_{\text{a,CT}}$ in E_1 remains a high value of 63.40 kJ mol^{-1} compared to 57.22 kJ mol^{-1} in E_2 , implying a more difficult process of Li^+ getting electrons from graphite, indicating the inorganic layer of SEI formed in E_1 wraps the graphite more completely.²⁹ Such an SEI in E_1 generates efficient protection of graphite to the electrolyte as well as fast Li^+ diffusion, leading to high interfacial stability during cycling and low resistance under low temperature.

Electrochemical Performance of 200 mAh NCM532||Graphite Pouch Cells. Inspired by the robust and low-

resistance interface of the graphite electrode in E_1 , 200 mAh NCM532||Gr pouch cells were tested to evaluate the electrochemical performance. Self-discharge properties were compared using leakage current and self-discharge tests, as shown in Figures 6a,b and S14. Pouch cells with E_1 exhibit the most stable interface with the lowest leakage-current value of 0.03612 mA and high cell voltage of 4.2423 V after 120 h of self-discharge test compared to E_0 and E_2 (Table S5). The results reflect the strong inhibitory effect of SEI formed in E_1 on the parasitic reaction of electrolyte or impurity, which is supported by the dense inorganic-rich SEI protecting the graphite borders completely. The low-temperature discharge capacity was evaluated, where the pouch cells were charged at 25°C and then discharged at various low temperatures (Figure 6c–f and Figure S16a). The pouch cell with electrolytes containing LiBF_4 exhibits much higher discharge capacities from 0 to -40°C . Specifically, the pouch cell with E_1 exhibits a highest capacity retention of 71.3% at -20°C in comparison with other pouch cells (69.1% and 64.4% for E_0 and E_2 , respectively). Detailed discharge potential curves and corresponding dQ/dV curves (Figure S16b,c) verify the fastest interfacial kinetics of pouch cells with E_1 , where E_1 shows the largest reaction voltage of 3.86 and 3.41 V compared to 3.76 and 3.33 V and 3.83 and 3.38 V for E_0 and E_2 , respectively. During cycling under -20°C , pouch cells with LiBF_4 also exhibit enhanced reversible capacity of 118.25 and 103.56 mAh under 0.1C for E_1 and E_2 , respectively, proving the advantage of reduced charge transfer resistance by BF_4^- . However, the pouch cells with E_2 show rapid capacity decay after 5 cycles, indicating the poor interfacial stability originating from the thick and inefficiently protective SEI. Encouragingly, pouch cells with E_1 achieve high capacity and stable cycling simultaneously, proving the SEI formed through enhanced LiPF_6 decomposition offset the poor passivation property of

LiBF₄. The pouch cells were disassembled after cycling at −20 °C, and the morphology of anodes and separators as well as the generated gases were compared (Figures S17 and S18). The results indicate the capacity decay is caused by solvent decomposition on anode surface, which further demonstrates a robust interface was constructed on graphite in the electrolyte with trace amount of LiBF₄ to achieve less solvent reduction for more stable cycling. Importantly, cycling performance under high temperature of 50 °C further verifies the improved interfacial stability of the pouch cell with E₁, as shown in Figure S19.^{46,47} Specifically, the pouch cell with E₁ retained 99.6% capacity after 53 cycles under 200 mA (1 h charge/discharge), while pouch cells with E₀ and E₂ show fast capacity decay with capacity retention of 97.8% and 97.0%. Combined with the discussion about solvation structure as well as interfacial composition and kinetics, it is demonstrated that regulating the solvation structure by rational adjustment of the strong interaction between Li⁺ and LiBF₄ could realize enhanced LiPF₆ decomposition for a robust LiF-rich interface. Such blended-lithium-salt electrolytes offer a collaborative route of anions on interface construction to achieve high capacity and stability simultaneously for low-temperature LIB pouch cells.

In conclusion, the contradictory effect of LiBF₄ on poor interfacial stability and low interfacial resistance was investigated, which effectively enhances the interfacial stability of LiBF₄-contained electrolytes by tuning the solvation structure. Rationally adjusting the amount of LiBF₄ could not only regulate the solvation structure but also tune the composition and morphology of the SEI. Due to strong solvation affinity of BF₄[−] with Li⁺, electrolytes with more than 0.1 M LiBF₄ forms an interface with more BF₄[−] decomposition products, which is not dense enough to inhibit continuous solvent decomposition, leading to a thick and unstable SEI. In comparison, the weakened interaction between BF₄[−] and Li⁺ by adding trace LiBF₄ (0.05 M) enables enhanced interaction between PF₆[−] and Li⁺ to form a dense and LiF-rich interface by more PF₆[−] decomposition, which inhibits solvent decomposition, forming a thin, robust, and low-resistance SEI. Moreover, the interaction between EC and Li⁺ is also weakened for low charge transfer resistance. Benefiting from the fully covered thin inorganic layer, low resistance for Li⁺ diffusion in SEI and fast desolvation process, electrolytes with trace LiBF₄ enables the 200 mAh NMCS32||Gr pouch cell to achieve less gas formation, low self-discharge, long-lived cycling, and large low-temperature capacity, simultaneously. This study provides a new understanding of solvation structure tuning by regulating interactions using various LiBF₄ amounts and also a method for constructing a stable interface for long-lived, low-temperature lithium-ion batteries.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenerylett.4c01616>.

Experimental and computational methods, NMR spectra of electrolytes under −20 °C, supplementary TEM images of SEIs, EIS spectra, optical images of anodes and separators obtained from cycled pouch cells, gas analysis, and high-temperature cycling stability test (PDF)

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Notes

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